

Week 4 Progress Report

Name: Blake Simpson Week: 4 (Midpoint) Track: Track 3 — Agent

What I did this week

- Added 3rd job source(Jobicy, ~100 remote jobs) as there still werent enough jobs being returned from the queries.
- Rewrote the ranking to match on title/tag instead of description which increased recall on the test queries from 38% to 63%.
- Built out email alerts, has existing jobs marked so that duplicate emails arent sent.
- Use score stability script to measure how much Claude's scores drift between identical runs; used this to lower the alert threshold to 8 from 9.
- Verified end to end flow with my real resume and I get email alerts when this runs

What worked

- The full pipeline worked on real input on the first try: my PDF résumé extracted cleanly, the agent derived sensible queries from it, and the SMTP send worked immediately (including the alert dedup correctly staying silent on the second run).

What Failed or Surprised me

- Remote's API is still broken two weeks later, re-verified that different search queries return byte-identical results from an ~18-hour-old CDN cache. The multi-source design absorbed this, but I've stopped expecting it to recover.
- The python-backend job test still finds 0 good matches even with the bigger corpus. It looked like a scoring failure but it's a coverage gap: today's feeds genuinely contain no Python-backend roles, and Claude scoring the mismatches low is correct behavior.
- Score drift is real but not very impactful. The rankings are stable (the best job stays on top) but absolute scores wobble ~1 point. The top match scored 8 one run and 9 the next. With a hard threshold of 9 it would have alerted only on lucky days, which is why the threshold is now 8.

What I learned

- Rank stability and score stability are different properties. The ranked list needs the first, and only the alert threshold needs the second, so instead of an expensive fix (multi-pass scoring, finer scales), moving the threshold off the drift boundary was enough, and the alert dedup makes the looser bar nearly free.
- When an agent produces bad output, check the data layer before blaming the model. Both "failures" this week (DevOps in Week 3, python-backend now) were corpus gaps that Claude was scoring correctly.

Evidence of Progress

- Midpoint doc with before/after tables and failure analysis: docs/week-4-midpoint.md
- Week 3 vs Week 4 baseline outputs, same 5 test tasks: docs/baseline_outputs_week3.md vs docs/baseline_outputs.md
- New code: src/job_scout/jobicy.py , src/job_scout/alerts.py , scripts/score_stability.py
- Commit: <https://github.com/EXC3ll3NTrhyTHM/agent-workflow/commit/1d92d15>
- Sample (my real résumé, live run — the agent self-corrected after round 1 and the dedup suppressed a repeat alert):

```
round 1/3: searching the job corpus for 'AI engineer Java'...
  scored [claude]: 2 good match(es) >= 7.0 so far (best 9.0)
  not enough good matches – deriving a fresh query (Claude, ~10-30s)...
round 2/3: searching the job corpus for 'llm engineer python'...
  scored [claude]: 3 good match(es) >= 7.0 so far (best 9.0)
  target of 3 good matches reached – stopping early
...
1 alert-worthy (>= 8.5):
  9.0 Senior AI Engineer Architect @ Lemon.io
All alert-worthy postings were already alerted on – no email.
```

Plan for Next week

- Schedule the daily cron job
- Start building the Week 6 eval harness: 20 test tasks (résumé profiles + expected relevance criteria) and a scoring rubric.

- Watch alert quality at threshold 8; if weak matches start emailing, try anchored score-band definitions in the scoring prompt before anything heavier.

Blockers

- None